

Homework 2 — Data Definition

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HW2

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script execution

the script `hw2.sql` drops all seven tables.

this is the second run of the script. All seven tables are dropped and recreated with no errors/warnings, confirming the script can be run repeatedly.

```
gknippel_db=> \i C:/CodeProjects/HW/CPSC321/hw2/hw2.sql
DROP TABLE
DROP TABLE
DROP TABLE
DROP TABLE
DROP TABLE
DROP TABLE
DROP TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
INSERT 0 5
INSERT 0 5
INSERT 0 5
INSERT 0 5
INSERT 0 6
INSERT 0 2
INSERT 0 3
INSERT 0 5
gknippel_db=>
```

table contents

```
gknippel_db=> SELECT * FROM airport;
```

air_id	air_name	city	state
GEG	Spokane International	Spokane	WA
SEA	Seattle-Tacoma International	Seattle	WA
LAX	Los Angeles International	Los Angeles	CA
DEN	Denver International	Denver	CO
JFK	John F. Kennedy International	New York	NY

(5 rows)

```
gknippel_db=> SELECT * FROM airline;
```

air_code	airline_name	air_id	founded_year
AS	Alaska Airlines	SEA	1932
WN	Southwest Airlines	LAX	1967
UA	United Airlines	DEN	1926
DL	Delta Air Lines	JFK	1925
AA	American Airlines	LAX	1930

(5 rows)

```
gknippel_db=> SELECT * FROM flight_route;
```

air_code	flight_num	dep_id	arr_id	distance
AS	122	GEG	SEA	224
AS	345	SEA	LAX	954
WN	122	LAX	GEG	1046
UA	588	DEN	JFK	1626
DL	701	JFK	LAX	2475

(5 rows)

```
gknippel_db=> SELECT * FROM passenger;
```

pass_id	first_name	last_name	email
1	Jack	Smith	jsmith@example.com
2	Alan	Cho	acho@example.com
3	Greyson	Knippel	gknippel@example.com
4	Maria	Delgado	mdelgado@example.com
5	Priya	Raman	praman@example.com

(5 rows)

```
gknippel_db=> SELECT * FROM booking;
```

pass_id	air_code	flight_num	travel_date	seat	price
1	AS	122	2026-10-02	14A	189.50
1	AS	122	2026-11-15	08C	210.00
2	AS	122	2026-10-02	14B	205.75
3	WN	122	2026-10-05		156.25
4	DL	701	2026-12-01	02A	645.00
5	WN	122	2026-10-05		161.00

(6 rows)

```
gknippel_db=> SELECT * FROM crew_member;
```

crew_id	first_name	last_name	hire_date	end_date	is_pilot	supervisor_id
1	Dana	Weiss	2015-03-01		t	
2	Omar	Diaz	2017-06-12		t	
3	Lena	Petrov	2019-01-20		f	1
4	Sam	Okafor	2020-09-05		t	2
5	Iris	Chen	2021-04-18	2026-02-28	f	2

(5 rows)

```
gknippel_db=> SELECT * FROM crew_assignment;
```

crew_id	air_code	flight_num	travel_date	shift_start	shift_end
1	AS	122	2026-10-02	2026-10-02 06:00:00	2026-10-02 14:30:00
2	AS	122	2026-10-02	2026-10-02 06:00:00	2026-10-02 14:30:00
1	AS	345	2026-10-03	2026-10-03 09:15:00	2026-10-03 18:45:00
4	DL	701	2026-12-01	2026-11-30 22:00:00	2026-12-01 06:00:00
5	UA	588	2026-10-10	2026-10-10 05:00:00	2026-10-10 20:00:00

(5 rows)

failing inserts

each statement below is commented out in the sql file.

1. duplicate primary key

duplicate primary key: 'GEG' is already the primary key of an existing airport row.

```
gknippel_db=> SELECT * FROM airport;
air_id | air_name | city | state
-----+-----+-----+-----
GEG    | Spokane International | Spokane | WA
SEA    | Seattle-Tacoma International | Seattle | WA
LAX    | Los Angeles International | Los Angeles | CA
DEN    | Denver International | Denver | CO
JFK    | John F. Kennedy International | New York | NY
(5 rows)

gknippel_db=> INSERT INTO airport (air_id, air_name, city, state) VALUES ('GEG', 'Duplicate Spokane', 'Spokane', 'WA');
ERROR: duplicate key value violates unique constraint "airport_pkey"
DETAIL: Key (air_id)=(GEG) already exists.
gknippel_db=>
```

2. foreign-key violation

foreign key violating: hub airport 'XXX' does not exist in airport.

```
gknippel_db=> SELECT * FROM airline;
air_code | airline_name | air_id | founded_year
-----+-----+-----+-----
AS       | Alaska Airlines | SEA | 1932
WN       | Southwest Airlines | LAX | 1967
UA       | United Airlines | DEN | 1926
DL       | Delta Air Lines | JFK | 1925
AA       | American Airlines | LAX | 1930
(5 rows)

gknippel_db=> INSERT INTO airline (air_code, airline_name, air_id, founded_year) VALUES ('B6', 'JetBlue Airways', 'XXX', 2000);
ERROR: insert or update on table "airline" violates foreign key constraint "airline_air_id_fkey"
DETAIL: Key (air_id)=(XXX) is not present in table "airport".
```

3. NOT NULL violation

non null violation: email cant be NOT NULL.

```
gknippel_db=> SELECT * FROM passenger;
pass_id | first_name | last_name | email
-----+-----+-----+-----
1       | Jack      | Smith    | jsmith@example.com
2       | Alan      | Cho      | acho@example.com
3       | Greyson   | Knippel  | gknippel@example.com
4       | Maria     | Delgado  | mdelgado@example.com
5       | Priya     | Raman    | praman@example.com
(5 rows)

gknippel_db=> INSERT INTO passenger (pass_id, first_name, last_name, email) VALUES (6, 'Nina', 'Ortiz', NULL);
ERROR: null value in column "email" of relation "passenger" violates not-null constraint
DETAIL: Failing row contains (6, Nina, Ortiz, null).
```

4. CHECK violation - departure equals arrival

check violation: departure and arrival airports must be different.

```
gknippel_db=> SELECT * FROM flight_route;
air_code | flight_num | dep_id | arr_id | distance
-----+-----+-----+-----+-----
AS      |      122 | GEG    | SEA    |      224
AS      |      345 | SEA    | LAX    |      954
WN      |      122 | LAX    | GEG    |     1046
UA      |      588 | DEN    | JFK    |     1626
DL      |      701 | JFK    | LAX    |     2475
(5 rows)

gknippel_db=> INSERT INTO flight_route (air_code, flight_num, dep_id, arr_id, distance) VALUES ('AS', 999, 'GEG',
'GEG', 100);
ERROR:  new row for relation "flight_route" violates check constraint "flight_route_check"
DETAIL:  Failing row contains (AS, 999, GEG, GEG, 100).
```

5. CHECK - negative price

check violation: booking price cant be negative.

```
gknippel_db=> SELECT * FROM booking;
pass_id | air_code | flight_num | travel_date | seat | price
-----+-----+-----+-----+-----+-----
1       | AS      |      122 | 2026-10-02 | 14A | 189.50
1       | AS      |      122 | 2026-11-15 | 08C | 210.00
2       | AS      |      122 | 2026-10-02 | 14B | 205.75
3       | WN      |      122 | 2026-10-05 |     | 156.25
4       | DL      |      701 | 2026-12-01 | 02A | 645.00
5       | WN      |      122 | 2026-10-05 |     | 161.00
(6 rows)

gknippel_db=> INSERT INTO booking (pass_id, air_code, flight_num, travel_date, seat, price) VALUES (2, 'DL', 701,
'2026-12-05', '11C', -50.00);
ERROR:  new row for relation "booking" violates check constraint "booking_price_check"
DETAIL:  Failing row contains (2, DL, 701, 2026-12-05, 11C, -50.00).
gknippel_db=>
```

6. UNIQUE violation - duplicate email

unique violation: (not the primary key): email is a candidate key, and this address already belongs to passenger 1.

```
gknippel_db=> SELECT * FROM passenger;
pass_id | first_name | last_name | email
-----+-----+-----+-----
1       | Jack      | Smith    | jsmith@example.com
2       | Alan      | Cho      | acho@example.com
3       | Greyson   | Knippel  | gknippel@example.com
4       | Maria     | Delgado  | mdelgado@example.com
5       | Priya     | Raman    | praman@example.com
(5 rows)

gknippel_db=> INSERT INTO passenger (pass_id, first_name, last_name, email) VALUES (7, 'Nate', 'Boyd', 'jsmith@ex
ample.com');
ERROR:  duplicate key value violates unique constraint "passenger_email_key"
DETAIL:  Key (email)=(jsmith@example.com) already exists.
gknippel_db=>
```

7. UNIQUE violation - seat already held

unique violation: (not the primary key): seat 14A on AS 122 for 2026-10-02 is already taken by another passenger.

```

gknippel_db=> SELECT * FROM booking;

```

pass_id	air_code	flight_num	travel_date	seat	price
1	AS	122	2026-10-02	14A	189.50
1	AS	122	2026-11-15	08C	210.00
2	AS	122	2026-10-02	14B	205.75
3	WN	122	2026-10-05		156.25
4	DL	701	2026-12-01	02A	645.00
5	WN	122	2026-10-05		161.00

```

(6 rows)

gknippel_db=> INSERT INTO booking (pass_id, air_code, flight_num, travel_date, seat, price) VALUES (3, 'AS', 122,
'2026-10-02', '14A', 199.00);
ERROR:  duplicate key value violates unique constraint "booking_air_code_flight_num_travel_date_seat_key"
DETAIL:  Key (air_code, flight_num, travel_date, seat)=(AS, 122, 2026-10-02, 14A) already exists.
gknippel_db=>

```

extra notes on the design

this implementation follows the part A design from hw1 without drastic change. all the tables, primary keys, and foreign keys are the same; the work in this assignment was mostly the data types and constraints.